Quasicrystals



MO-201 Materials Science
Odd Sem 2026-27

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What are Quasicrystals?

- Quasicrystals have properties of both crystalline structures, like diamonds, and amorphous structures — structures without an atomically definite form — like glass; like crystals, the atoms of quasicrystals are organized into patterns; like amorphous solids, those patterns do not repeat

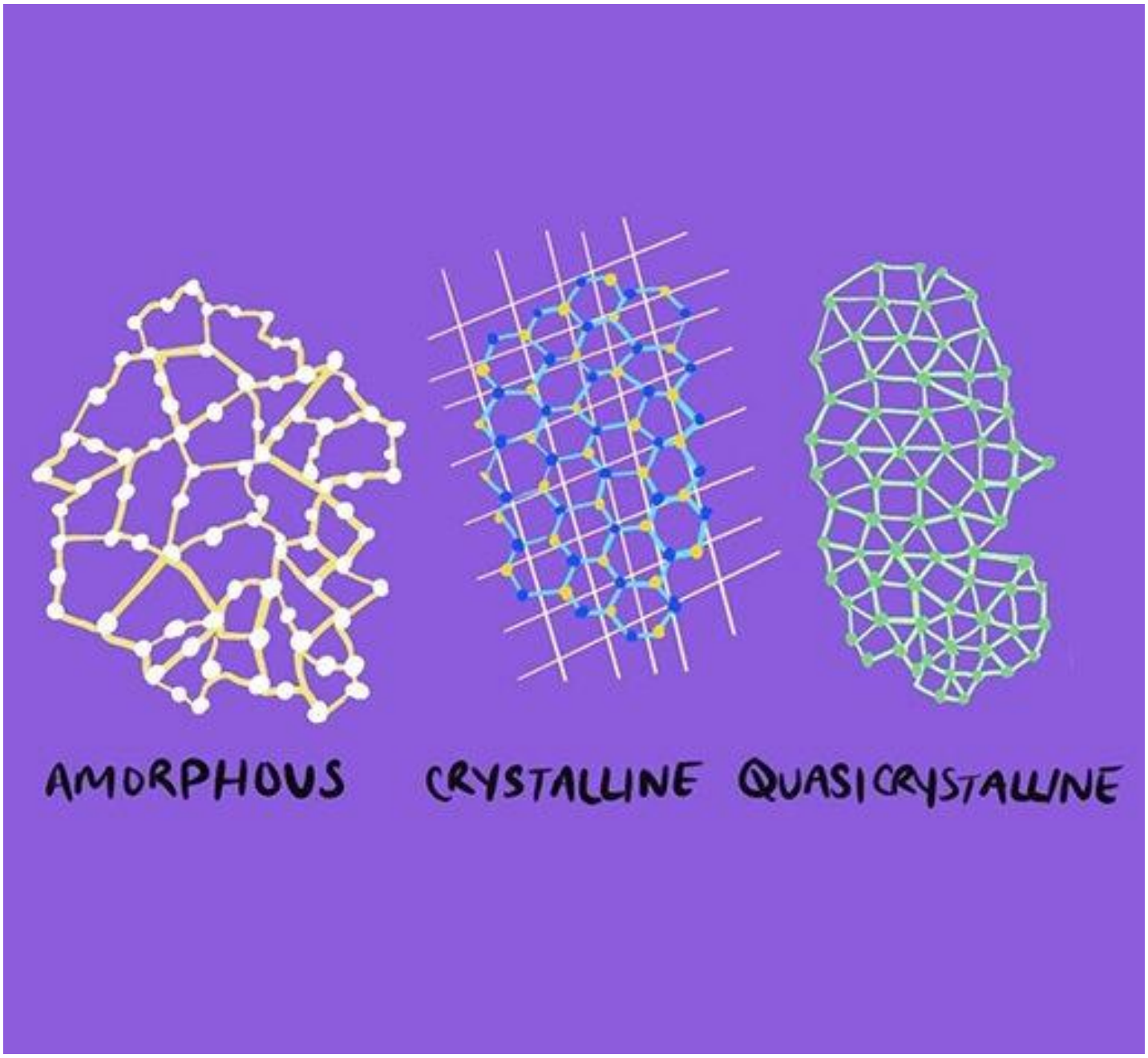
- Regular crystals exhibit two-fold, three-fold, four-fold, or six-fold symmetry at the atomic level; quasicrystals can additionally exhibit five-fold, ten-fold, or twelve-fold symmetry

- Scientist Dan Shechtman discovered quasicrystals in 1982 while researching Al and Mn at the National Institute of Standards and Technology in Maryland, US

- Shechtman and his team melted the two metals together and rapidly cooled them back to a solid to create a mixture of metals called an alloy

- After viewing the alloy under an electron microscope, Shechtman noticed it exhibited ten-fold symmetry, writing “10-fold???” in his journal notes. At that time, it was established that this does not happen with crystalline structures like metals

- This is the first known record of quasicrystals; Shechtman later won the Nobel Prize in Chemistry in 2011 for his discovery


AMORPHOUS
CRYSTALLINE
QUASICRYSTALLINE

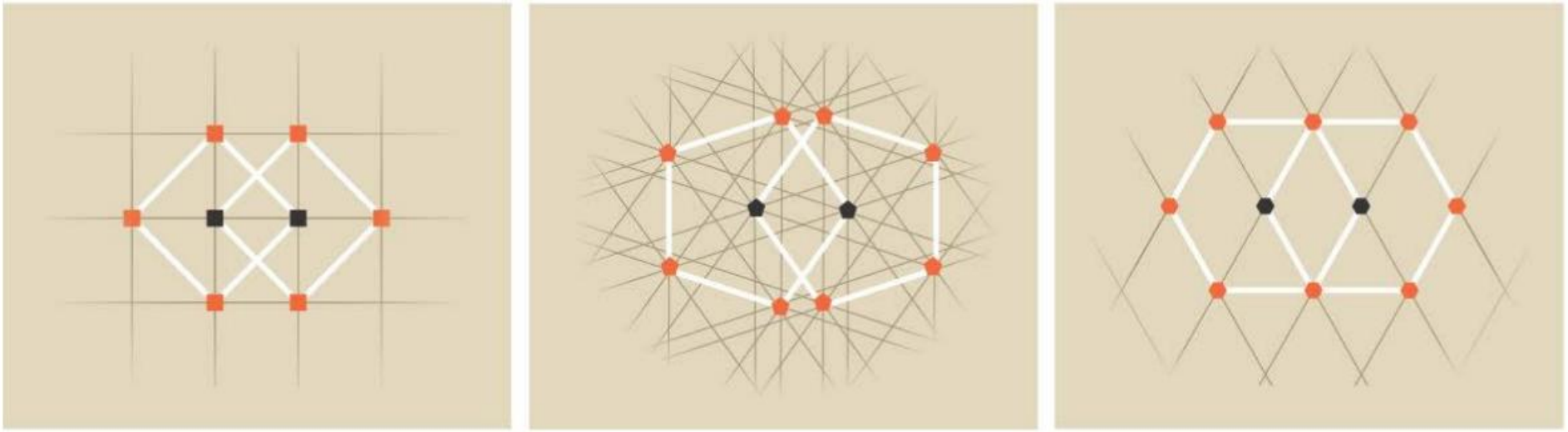


Figure 1. Two 4-fold (left) or 6-fold (right) axes of rotation generate new rotational axes at the same distance of separation as in the original pair. Repeating the procedure yields periodicity. For the pair of 5-fold axes (centre), the procedure instead generates a new, shorter distance. An iterative procedure will thus fill the plane densely with 5-fold axes, and no periodicity will result.

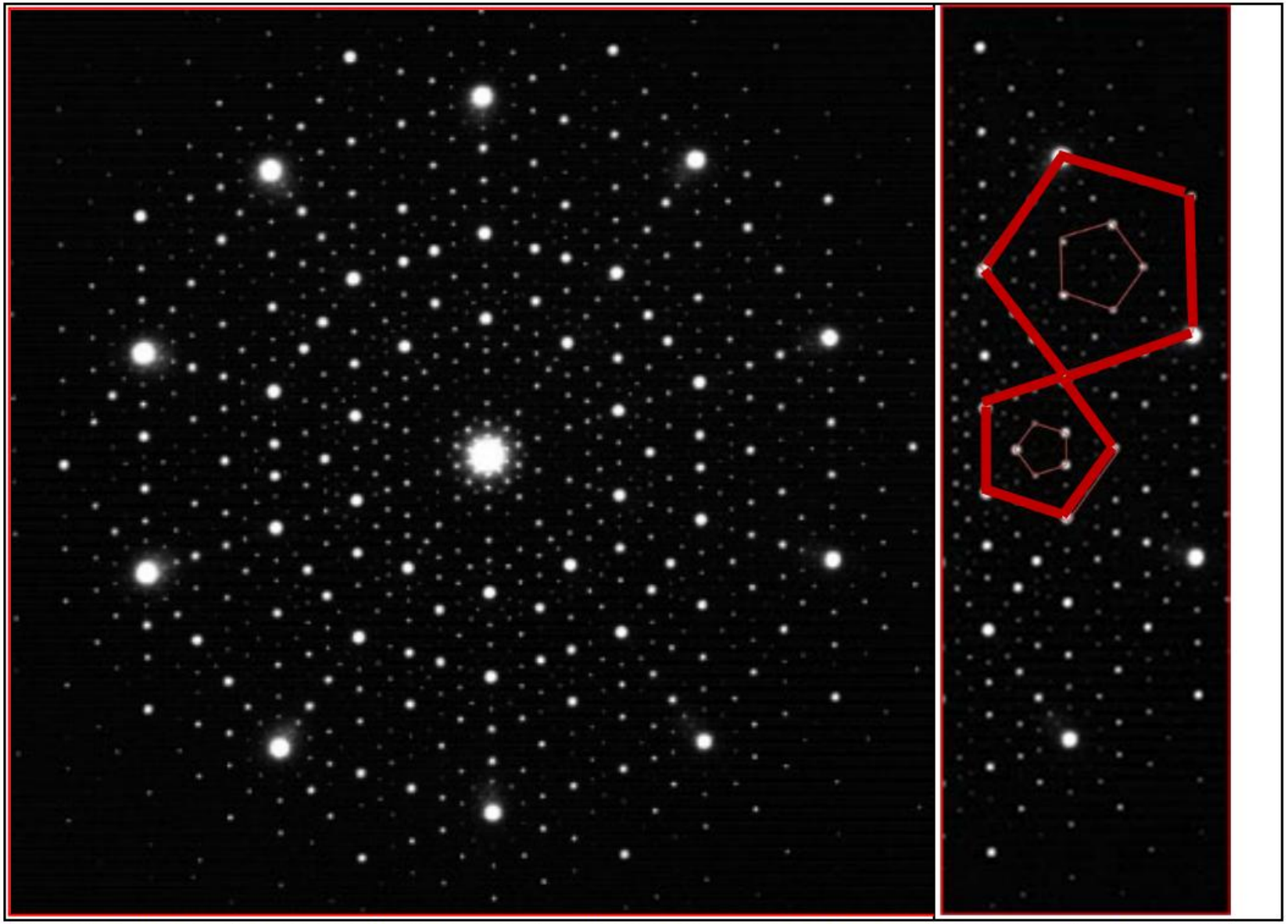


Figure 2. Electron diffraction pattern from an icosahedral quasicrystal. Note the presence of perfect pentagons highlighted in the diagram to the right. The linear scale between pentagons is τ, and the scale between a pentagon inscribed in another pentagon is τ^2. One of the great challenges of quasicrystal structure work is also apparent in this image. The intensity distribution of the diffraction pattern varies over many orders of magnitude, even in electron diffraction, and measuring the crucial weak reflections in an X-ray diffraction pattern within a reasonable time frame has only become possible with modern area detectors that were not around at the time of discovery.

What are Quasicrystals (contd.)?

- A quasicrystal is a material that exhibits long-range order in a diffraction experiment and yet does not have translational periodicity

- In a seminal paper published in November 1984 (Shechtman et al.), rapidly solidified alloys of Al with 10–14% Mn were shown by means of electron diffraction to possess icosahedral symmetry in combination with long-range order

- A very important factor for the successful determination of quasicrystal structure has been the discovery of stable quasicrystals; stable quasicrystals may be grown to considerable size and exhibit the typical features of well-ordered crystalline phases

- The first stable icosahedral quasicrystals were synthesized as early as 1987 in the ternary system Fe-Cu-Al

- A breakthrough came with the discovery of a binary stable icosahedral quasicrystal in 2000, using Ca-Cd and Yb-Cd

- Quasicrystals are solids made of conductive metals like aluminum, copper, and iron and are used in many practical items, like non-stick frying pans, needles, and LED lights

References

- A journey through the history of quasicrystals – The Varsity By Chris Zdravko — Published March 3, 2024

- nobelprize.org, Scientific Background on the Nobel Prize in Chemistry 2011, The Discovery of Quasicrystals